



Netflix Stock Price Prediction

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Course

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Abstract

- The primary objective of this report is to predict Netflix Inc.'s (NFLX) stock prices using an Exponential Moving Average (EMA) model, the attributes included in the model are Date, Open, Close, High, Low, Volume, and Adj Close.
- Data Collected from 2018 to 2022 will be divided into two parts, the training set will be defined as 80% and the testing set is 20% of the total dataset, and it will be used to predict NFLX stock prices.

Introduction

- The stock market is known for being volatile, dynamic, and nonlinear. Accurate stock price prediction is extremely challenging because of multiple (macro and micro) factors, of which only a few can be quantified, so rather than focusing on matching the actual values with high precision, we try to get a probabilistic estimate of what the market “could” look like.
- The data set that will be used in action is collected from `finance.yahoo[1]`, the collected data spans from February 1st, 2018, to December 30th, 2022, with a daily step interval, the dataset features 5 attributes:
 1. Open: the price from the first transaction of a business day.
 2. High: the highest price at which a stock is traded during the business day.
 3. Low: the lowest price at which a stock is traded during the business day.
 4. Close: the last price anyone paid for a share of stock during a business day.
 5. Adj Close: the closing price after adjustments for all applicable splits and dividend distributions.
 6. Volume: the number of shares traded in a stock or contracts traded in futures or options (indicates market strength).

Model

- The chosen model for this project is the moving average (MA), because of its ability to smooth out random and unpredicted data it is generally used as an ML model in predicting time series problems such as stocks in the market.
- Commonly used periods are 20-day, 50-day, and 200-day MA for short-term, medium-term, and long-term investment respectively, in this case, medium-term MA will be used since it is better a trend indicator in terms of short to medium movements.
- The 2 types of MA most preferred by financial analysts are Simple MA and Exponential MA, so which is better in this case?
- Firstly, SMA calculates the average of a range of stock prices over a specific number of periods in that range. $SMA = \frac{P_1 + P_2 + \dots + P_n}{N}$ where P_n = the stock price at time point n, N = the number of time points (50 in this case).
- While SMA assigns equal weights to all data points, EMA applies higher weights to recent prices, the magnitude of the weighting factor depends on the number of time periods. $EMA = P_t \times k + EMA \times (1 - k)$ where P_t = the price at time point t, EMA_{t-1} = EMA at time point t-1, N = number of time points in EMA, and weighting factor $k = 2/(N+1)$.
- Both of the formulas show that EMA is more responsive to price changes, which makes it useful for short-term trading making it work well in this case.
- Regarding evaluation metrics, the Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error % (MAPE) will be used because both are useful measures of forecast accuracy.

- RMSE gives the differences between predicted and true values rooted.

$$RMSE = \sqrt{\frac{1}{N} + \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

- MAPE (%) measures the difference relative to the true values.

$$MAPE = \frac{1}{N} \times \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

where N = the number of time points, y_i = the value, $\hat{y}_i$ = the predicted value.

Conclusion

- In conclusion, this report is about using a machine learning model (EMA) to predict the price of NFLX stocks based on a 50-day average. The evaluation will be done using two metrics, MAPE and RMSE. The prediction process will use daily data from the NTFLX stock dataset[1], covering the years 2018 to 2022.

References

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3. <https://neptune.ai/blog/predicting-stock-prices-using-machine-learning>
4. Agrawal, J. G., Chourasia, V., & Mittra, A. (2013). State-of-the-art in stock prediction techniques. International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, 2(4), 1360-1366.